

Regularization - Numerical Questions

Q1. L1 Regularization

Given: $w_1 = 2$, $w_2 = -3$, $\lambda = 0.5$.

Original loss = 10.

Calculate the regularized loss using L1 regularization.

Formula: $L_{\text{total}} = L + \lambda \sum |w_i|$

Q2. L2 Regularization

Given: $w_1 = 2$, $w_2 = -3$, $\lambda = 0.1$.

Original loss = 5.

Calculate the regularized loss using L2 regularization.

Formula: $L_{\text{total}} = L + \lambda \sum w_i^2$

Q3. L1 vs L2

Given: $w_1 = 3$, $w_2 = -4$, $w_3 = 2$, $\lambda = 0.2$.

Calculate:

1. L1 penalty
2. L2 penalty
3. L1 regularized loss if original loss = 8
4. L2 regularized loss if original loss = 8

Q4. Ridge Regression

Given: $w_1 = 4$, $w_2 = 3$.

MSE = 6, $\lambda = 0.5$.

Calculate the Ridge regularized loss.

Q5. Lasso Regression

Given: $w_1 = 5$, $w_2 = -2$, $w_3 = 1$.

Loss = 8, $\lambda = 0.4$.

Calculate the Lasso regularized loss.

Q6. L2 Weight Update

Given: $w = 5$, $\eta = 0.1$, $\partial L / \partial w = 2$, $\lambda = 0.5$.

Calculate the new weight using:

$w_{\text{new}} = w - \eta(\partial L / \partial w + 2\lambda w)$

Q7. L1 Weight Update

Given: $w = 4$, $\eta = 0.1$, $\partial L / \partial w = 3$, $\lambda = 0.5$.

Calculate the new weight using L1 regularization.

Q8. Ridge with Multiple Weights

Given: $w_1 = 2$, $w_2 = -5$, $w_3 = 3$.

MSE = 12, $\lambda = 0.2$.

Calculate the total Ridge loss.

Q9. Lasso with Multiple Weights

Given: $w_1 = -4$, $w_2 = 6$, $w_3 = -2$.

MSE = 10, $\lambda = 0.3$.

Calculate the total Lasso loss.

Q10. Elastic Net

Given: $w_1 = 2$, $w_2 = -3$.

MSE = 7, $\lambda = 0.5$, $\alpha = 0.6$.

Calculate the Elastic Net regularized loss using:

$$L_{\text{total}} = L + \lambda[\alpha \sum |w_i| + (1 - \alpha) \sum w_i^2]$$

Regularization & Cost Function - 5 Numerical Questions

Q1. L2 Regularized Cost Function

Given $m = 5$, prediction errors are 2, -1, 3, -2, and 1. The model has weights $\theta_1 = 2$ and $\theta_2 = -3$, with $\lambda = 0.4$.

Calculate:

1. The original cost function $J(\theta)$ using $J(\theta) = (1/2m) \sum (h_{\theta}(x^{(i)}) - y^{(i)})^2$.
2. The L2 regularization term $\lambda/(2m) \sum \theta_j^2$.
3. The total regularized cost.

Q2. L1 Regularized Cost Function

Given $m = 4$, prediction errors are 1, -2, 2, and -1. The weights are $\theta_1 = 3$ and $\theta_2 = -4$, with $\lambda = 0.5$.

Calculate:

1. The original cost function $J(\theta)$.
2. The L1 regularization term $\lambda/(2m) \sum |\theta_j|$.
3. The total regularized cost.

Q3. Compare Regularized and Unregularized Cost

For $m = 5$, the squared prediction errors are 4, 1, 9, 4, and 1. The weights are $\theta_1 = 2$, $\theta_2 = -2$, and $\theta_3 = 1$. Let $\lambda = 0.5$.

Calculate:

1. The unregularized cost $J(\theta)$.
2. The L2 regularization penalty using $\lambda/(2m) \sum \theta_j^2$.
3. The regularized cost.
4. The increase in cost due to regularization.

Q4. Ridge Cost Function

A linear regression model has $m = 10$ observations. The sum of squared errors is 80. The model weights are $\theta_1 = 3$, $\theta_2 = -4$, and $\theta_3 = 2$. Use $\lambda = 0.2$.

Calculate the Ridge cost using:

$$J_{\text{Ridge}}(\theta) = \text{SSE}/(2m) + \lambda/(2m) \sum \theta_j^2.$$

Q5. Elastic Net Cost Function

Given $m = 6$, the sum of squared errors is 48. The weights are $\theta_1 = 2$ and $\theta_2 = -3$. Let $\lambda = 0.6$ and $\alpha = 0.7$.

Calculate the Elastic Net cost using:

$$J(\theta) = \text{SSE}/(2m) + \lambda/(2m) [\alpha \sum |\theta_j| + (1 - \alpha) \sum \theta_j^2].$$